



Cloud Computing

— Introduction —

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What Is The Buzz About

- Hot topic for large computing vendors (Amazon, Google, Microsoft etc.) **only from mid-2008**
- The concept behind it is **not completely new:**
 - Access the web-based services through Internet
 - Management of computing services is outsourced to greater extents and in multiple dimensions
 - Computing facilities are being made available as services offered by reputed vendors
- *Cloud computing has evolved over the years and has changed the way people live and do business over the World Wide Web*

What Is The Buzz About

- The means for users to easily avail computing facilities whenever and wherever required
 - Can access any volume, large or small, of computing facilities in exchange for some nominal payment
- Not worry about
 - Setting up infrastructure,
 - Purchasing new equipment, or
 - Investing in the procurement of licensed software

What Is The Buzz About

- New model of computing which has become possible through integration of:
 - Advanced computing models,
 - Sophisticated web technologies, and
 - Modern network communication technologies (especially high-speed Internet).
- *Cloud computing is the delivery of computing services over the Internet.*

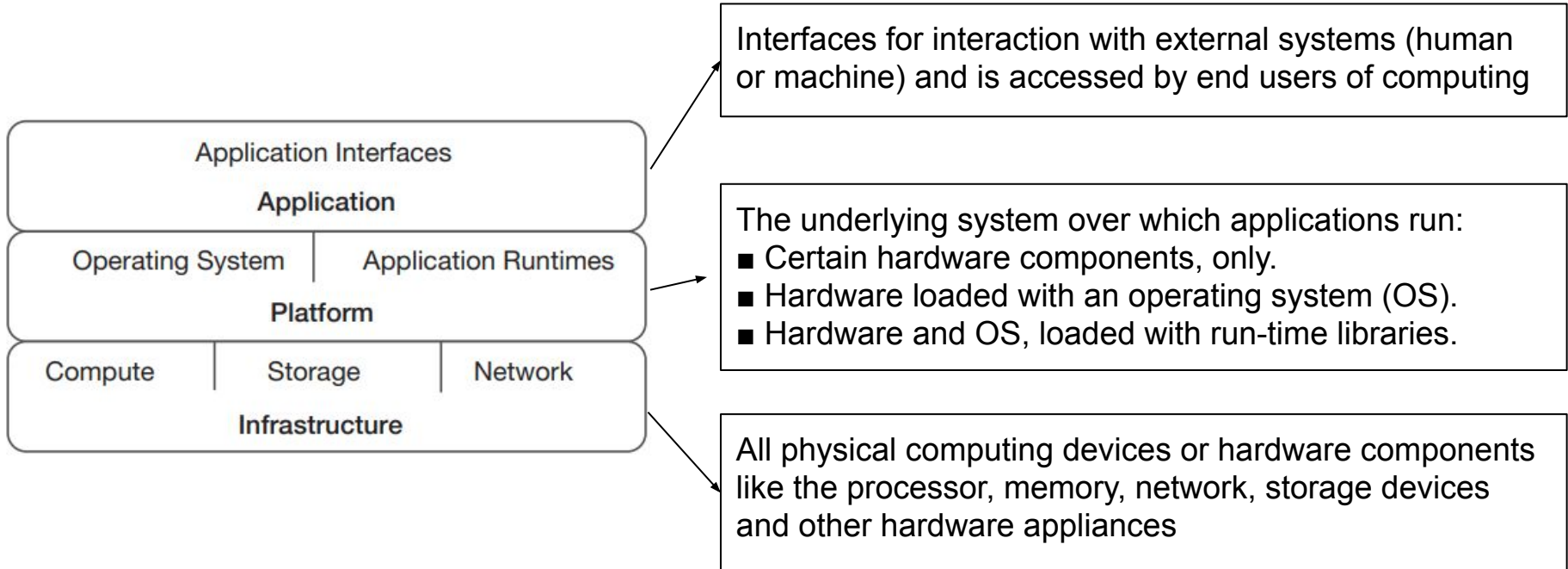
Limitations Of Traditional Computing Approaches

- Enterprise Perspective:
 - Huge initial investment for setting up computing infrastructure
 - Difficulty of managing the complex systems
 - Maintain a team of experts (system maintenance team)
 - Difficulty in regularly replacing outdated things with latest available technology
 - Capacity planning of computing resources
- Individual User's Perspective:
 - Initial necessities of procuring expensive physical system
 - Maintenance of the hardware and software

Is There Any Solution To These Worries?

- Doing it themselves (by deploying in-house team, in case of an organization)
- Outsourcing the responsibility to some third party who are well-trained for the job

Three Layers Of Computing



The End Of Traditional Computing

In traditional computing model:

- Subscribers of one computing layer cannot fully escape the responsibilities and difficulties of arranging or managing the underlying layer(s).
- Computing subscribers were always over-burdened with many additional difficulties and cost.

Computing as the Utility Service

- Three major aspects of computing which were represented in three-layered computing architecture are delivered as utility services
 - Infrastructure Service
 - Platform Service
 - Application or Software Service
- *Since, cloud resides at some remote location and does all the heavy computing tasks, the hardware and software demands on the local computers at users' side decrease.*

Who Provides the Service?

- Independent Software Vendors (ISV) generally provide the service.
 - Reputed computing/IT giants.
 - Owners or developers of the clouds and manage everything
- Users can use cloud services by payment on use basis,
 - Just like they pay monthly bill for electricity service

-> *profitable for both parties, the customers and the vendors*

Concerns

- Is this computing service (cloud computing) provided by others from some remote place reliable?
- What is the guarantee that the cloud service vendor will keep on providing reliable and uninterrupted service?
- Why should a user or enterprise trust a vendor when security or privacy of confidential information is involved?

Concerns

Adoption of cloud computing also requires entrusting another company (the cloud vendor) with subscriber's personal, official and confidential data

One thing is evident, the reliability and safeguards of cloud computing are no less if not better than the traditional computing system. Moreover, it comes with a host of other facilities and advantages.

Cloud computing signifies a major change in the approach how information is stored and applications can run.

Story Of A Similar Scenario: A Century Ago

- In the early twentieth century, before the advent of the electricity grid, industries used to build and run their own power generation plants just as many enterprises were to deploy and manage their own computing systems in traditional computing system



Influences Behind Cloud Service Adoption - Technological Influences

- Universal Network Connectivity
- High-Performance Computing
- Commoditization

Cloud computing has created the scope where consumers can rent a server on Internet and work from any place by signing-in to gain the access.

Cloud computing provides access to supercomputer-like power at a fraction of the traditional cost.

Influences Behind Cloud Service Adoption - Operational or Business Influences

- Low Cost Solution
- Outsourcing
- Speed or Responsiveness
- Automation
- Small Initial Investment
- Less Maintenance Cost
- Mobility

Summary

- Cloud computing has emerged as a new model of computing which promises to make the planet more smart
- There were many limitations in the traditional computing approaches. Those limitations used to overburden the consumers and enterprises in terms of budgetary expense and responsibility for maintaining various computing facilities.
- Cloud computing facilitates the delivery of computing services like any other utility service. Customers can use computing service just as electricity is consumed and pay bills at the end of the month

Summary

- The key advantage of this new model of computing is the access to any kind and any volume of computing facilities from anywhere and anytime.
- The major concern among the customers regarding the cloud computing service is about security and privacy of sensitive and personal information. But, with IT outsourcing, information security concerns existed in traditional computing approaches too.
- Reputed computing vendors have started providing cloud computing services. Their expertise in computing makes these services robust and their volume of business makes it cheaper for consumers.